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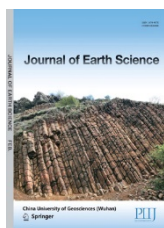


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Quantitative constraints on the exhumation and preservation of the giant Jinchuan Cu-Ni sulfide deposit

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ABSTRACT: Jinchuan deposit, the world's 3rd largest Cu-Ni sulfide magmatic deposit formed in the Neoproterozoic, has been developed as an open pit mine, indicating the impact of exhumation and erosion on the deposit. To evaluate the preservation of the Jinchuan deposit, this study compiles published thermochronological data (zircon (U-Th)/He, apatite (U-Th)/He, apatite fission track) and Monte Carlo inverse thermal modeling results, with the aim to reconstruct <300 Ma time-temperature of the Jinchuan deposit when the deposit was exhumed to zircon (U-Th)/He temperature range. Furthermore, we conduct 2D explicit finite difference inverse thermal modeling and the results suggest that near-surface exposure of the Jinchuan deposit most likely occurred during the mid-Cretaceous (ca. 90 Ma), which is corresponding to the timing when the Jinchuan deposit entered the apatite (U-Th)/He closure temperature range and is much earlier than a previously suspected Miocene exposure. A total cumulative exhumation thickness of 2.7~6.2 km and 1.8~3.7 km is calculated for time periods since initial exhumation (ca. 300 Ma) and surface exposure (ca. 90 Ma). It is noted that post-exposure erosion thickness is thicker than the current maximum thickness of the residual ore-bearing intrusion (1.2 km). Prolonged post-mineralization exhumation facilitates the shallow exhumation and ultimately leads to exposure, partly erosion and a degree of preservation with appreciable economic value for the ancient pre-Cambrian Cu-Ni sulfide Jinchuan deposit.

KEY WORDS: Jinchuan Cu-Ni sulfide deposit, thermochronology, exhumation, preservation.

0 INTRODUCTION

The Jinchuan magmatic Cu-Ni sulfide deposit, formed by the intrusion of high-Mg ultramafic magma into the Paleoproterozoic metamorphic rocks of the Longshoushan Complex at ca. 830 Ma,

is world's 3rd largest nickel resource (Duan, 2015; Song et al., 2009; Tang et al., 2009; Zhang et al., 2010). The ore-bearing intrusion is currently exposed at the surface, and open-pit mining has revealed evidence of post-emplacement exhumation and erosion, which may have compromised the preservation of metal reserves (England and Molnar, 1990). Previous studies indicate that the ore-bearing intrusion has undergone monotonic cooling since entering the range of zircon (U-Th)/He closure temperature at ca. 300 Ma (Cai et al., 2023; Tao et al., 2023; Tao et al., 2025, see 3.2 for details). However, the precise timing of its near-surface exposure and the extent of post-exposure erosion remain unclear. Lei et al. (2023) proposed a Miocene (ca. 15 Ma) exhumation event based on apatite fission-track data and inverse thermal modeling results (Monte Carlo method, Ketcham, 2005), and identified it as the last rapid cooling event of the Jinchuan deposit, which led to the near-surface exposure of the intrusion. Yet, whether this event marks the near-surface exposure and how much of the ore-bearing intrusion has been eroded are still unresolved questions.

Thermochronology is a group of isotopic dating methods used to constrain the timing of rock cooling or exhumation through the accumulation of daughter products from radioactive decay or fission in minerals (Fleischer and Price, 1964; Gleadow and Seiler, 2015; Zeitler et al., 1987). Common techniques include (U-Th)/He, fission track, and $^{40}\text{Ar}/^{39}\text{Ar}$ dating, each with its own thermal diffusion or annealing model and closure temperature range (Liu et al., 2024; Xia et al., 2025). When integrated, these systems enable the reconstruction of time-temperature or time-depth exhumation histories (Ding et al., 2018; Ge et al., 2013; Ju et al., 2023; Peng et al., 2018; Sauter et al., 2013; Shen et al., 2009; Shuster et al., 2011; Willett and Brandon, 2013). In recent years, growing attention has been paid to the use of thermochronological data and inverse thermal modeling to reconstruct the exhumation history and preservation of ore deposits, as exemplified by studies on gold deposits in the Jiaodong (Chen et al., 2023; Liu et al., 2017; Sun et al., 2017; Zhang, L. et al., 2017; Zhang et al., 2022) and Western Qinling regions (Peng et al., 2018; Yang et al., 2025; Yu et al., 2022).

This study reconstructs the thermal and exhumation history of the Jinchuan ore-bearing intrusion since ca. 300 Ma by compiling published low-temperature thermochronological data and two-dimensional explicit finite-difference inverse thermal modeling (Fu et al., 2010). The results are also used to quantify the post-exposure exhumation thickness, providing new insights into the preservation state of this world-class magmatic sulfide deposit.

1 GEOLOGICAL SETTING

1.1 Longshoushan Terrane

The Jinchuan deposit is situated in the Longshoushan Terrane, part of the NW-trending Longshoushan Thrust Belt (LSSTB) on the northeastern margin of the Tibetan Plateau (Li et al., 2022; Tao et al., 2025; Zeng et al., 2013). This terrane is bounded by the Southern and Northern Longshoushan Faults (S-LSSF and N-LSSF), which separate it from the Hexi Corridor to the south and the Chaoshui Depression to the north (Fig. 1a, Tang and Li, 1995; Zhang, B. H. et al., 2017). The region underwent multiple tectono-thermal episodes, including Early Paleozoic granitoid magmatism during the Qilian orogeny (Dong et al., 2021; Song et al., 2013) and Cenozoic crustal shortening linked to Tibetan Plateau uplift (Zhao, D. et al., 2023).

The basement of the Longshoushan Terrane consists of Neoarchean to Paleoproterozoic metamorphic units (Baijiazui and Tamazigou), overlain by Meso- to Neoproterozoic metasedimentary rocks (Gong et al., 2011; Tang et al., 2023; Xiu et al., 2004). A suite of Neoproterozoic (ca. 830 Ma, Zhang et al., 2010) mafic-ultramafic intrusions, including the host rocks of the Jinchuan Cu-Ni sulfide deposit, were emplaced in a tectonically active setting along the southwest edge of the Alax Block in the North China Craton (Fig. 1a, Li et al., 2005; Tang and Ren, 1988). Although the intrusions are spatially aligned with these faults in regional NW-trending fault systems, current evidence does not conclusively demonstrate that they were emplaced along major strike-slip faults. Instead, later deformation and multiple reactivation phases, particularly during the Paleozoic and Cenozoic, resulted in strike-slip dismemberment and dislocation of the original intrusions (Song et al., 2012; Tao et al., 2023). Subsequent mafic magmatism around 420 Ma, represented by the Xijing body and mafic dykes in the Jinchuan area, reflects a later pulse of mantle-derived activity (Duan et al., 2015), which was not directly involved in the formation of the Jinchuan deposit. This multi-phase magmatism and deformation history underscores the complex tectono-magmatic evolution of the Longshoushan region (Song et al., 2012; Zeng et al., 2013).

1.2 Jinchuan magmatic Cu-Ni sulfide deposit

The Jinchuan magmatic Cu-Ni sulfide deposit, discovered in 1959, is located in the eastern section of the LSSTB and adjacent to the N-LSSF (Fig. 1a). This deposit alleviated nickel and

platinum shortages in China (Li, 2022). The Jinchuan deposit has attracted attention because almost the entire ore-bearing intrusion is mineralized with a mineral content of up to 70% (Chai and Naldrett, 1992; Tang and Li, 1995), although the surface area exposed is only about 1.34 km².

The exposed strata in Jinchuan mining area are mainly Paleoproterozoic gneisses and marbles of the Baijiazui Unit, which underwent partial melting and reached upper amphibolite facies metamorphism (Lehmann et al., 2007; Liu et al., 2023). The Jinchuan ultramafic magma intruded this unit, with marbles to the north and migmatites to the south (Gong et al., 2012; Tang et al., 2023; Tao et al., 2023; Xiu et al., 2004). The deposit contains hundreds of ore bodies, mostly magmatic in origin (Yang et al., 2006), hosted in mafic and ultramafic intrusions. The main ultramafic lithologies are pyroxene-bearing peridotite, lherzolite, and plagioclase lherzolite (Fig. 1b, c).

The Jinchuan ore-bearing intrusion crystallized during the Neoproterozoic (ca. 830 Ma, Zhang et al., 2010) and was subsequently crosscut by mafic-ultramafic dykes. One such dyke yielded a zircon U-Pb age of 828 ± 3 Ma (Li et al., 2005), suggesting it is broadly coeval with the Jinchuan intrusion. Song et al. (2014) proposed that the mafic dykes represent the parental melt of the Jinchuan ultramafic intrusion. However, both geochemical and petrological evidence indicate that these dykes are more evolved than the basaltic parental magma of the intrusion (Li and Ripley, 2011; Li et al., 2005). Furthermore, zircon U-Pb dating of diabase dykes within the mining area yields an age of ca. 420 Ma (Duan et al., 2015), indicating multiple magmatic episodes.

The Jinchuan ore-bearing intrusion is cut by a series of post-emplacement NNE-trending faults (F8, F16, F23), dividing it into four segments from west to east: III, I, II, and IV (Fig. 1b). The main ore bodies (No. 1, No. 2, and No. 24), which account for over 90% of the deposit's total Ni-Cu-PGE reserves, are mainly hosted in segments II and I (Dai et al., 2015). These ore bodies extend downward, implying significant potential for deep exploration (Gao et al., 2020; Wang et al., 2019).

The Jinchuan intrusion lies near the unconformity between Paleoproterozoic basement and Meso-Neoproterozoic cover rocks (Lehmann et al., 2007), indicating it may have been close to the surface during Upper Proterozoic sedimentation (Tao et al., 2023). The unconformity between the Carboniferous and Neoproterozoic strata suggests the surface exposure of Neoproterozoic sequences (Fig. 2a, Tao et al., 2023). And on the south of Jinchuan mining area, Devonian siltstones are in direct fault contact with siliceous dolomites from Mesoproterozoic sequences (Fig. 2b, GSBGMR, 1995). Above all, considering the shallow emplacement of the Jinchuan ore-bearing

intrusion during Neoproterozoic, the intrusion was likely near the surface before Silurian period (Tao et al., 2023).

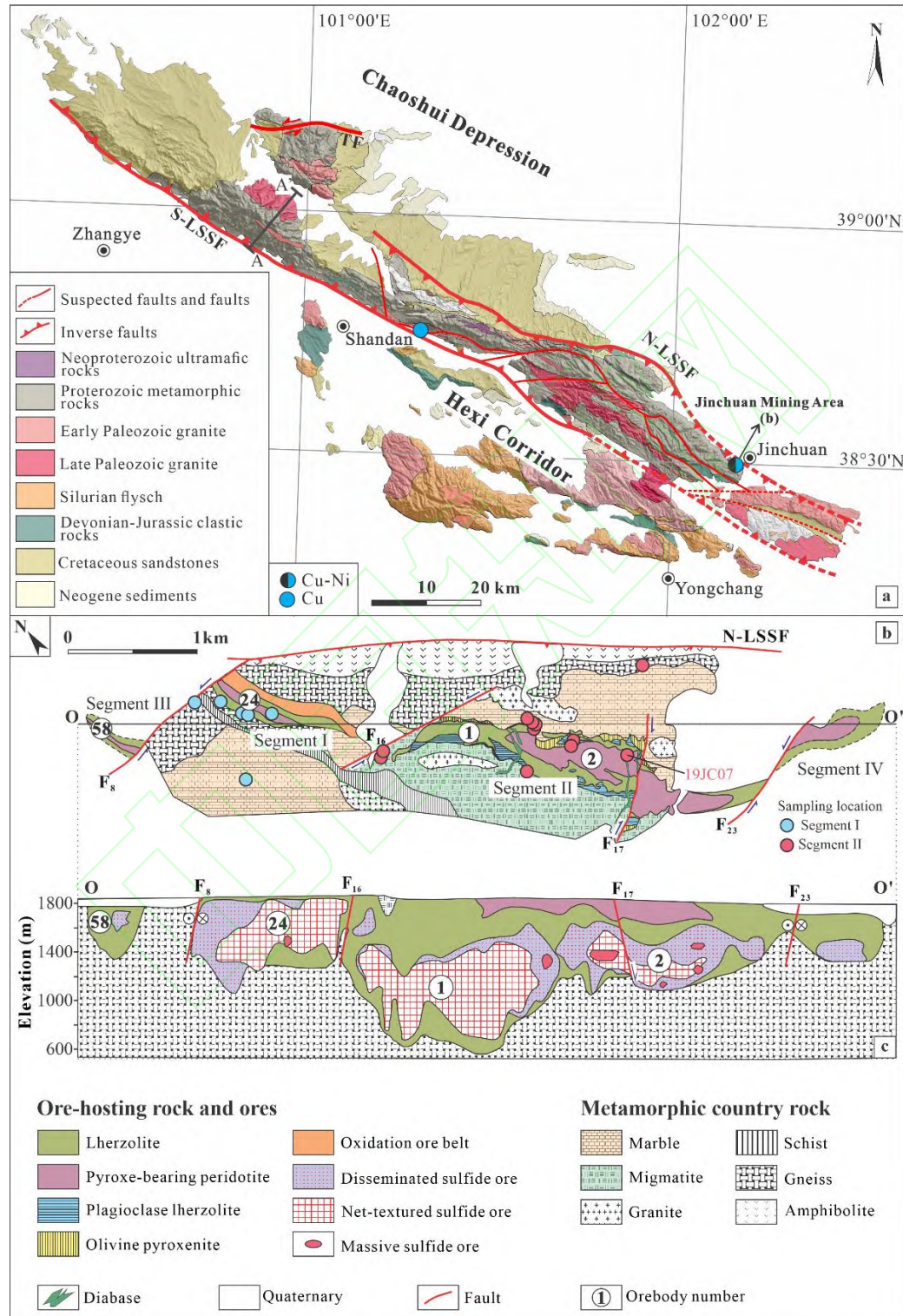


Figure 1. The geological map of the Longshoushan Terrane (a), geological map of Jinchuan mining area showing locations of samples in Table 1 (b), vertical projection map of Jinchuan ore deposits (c). TF: Taohualashan Fault.

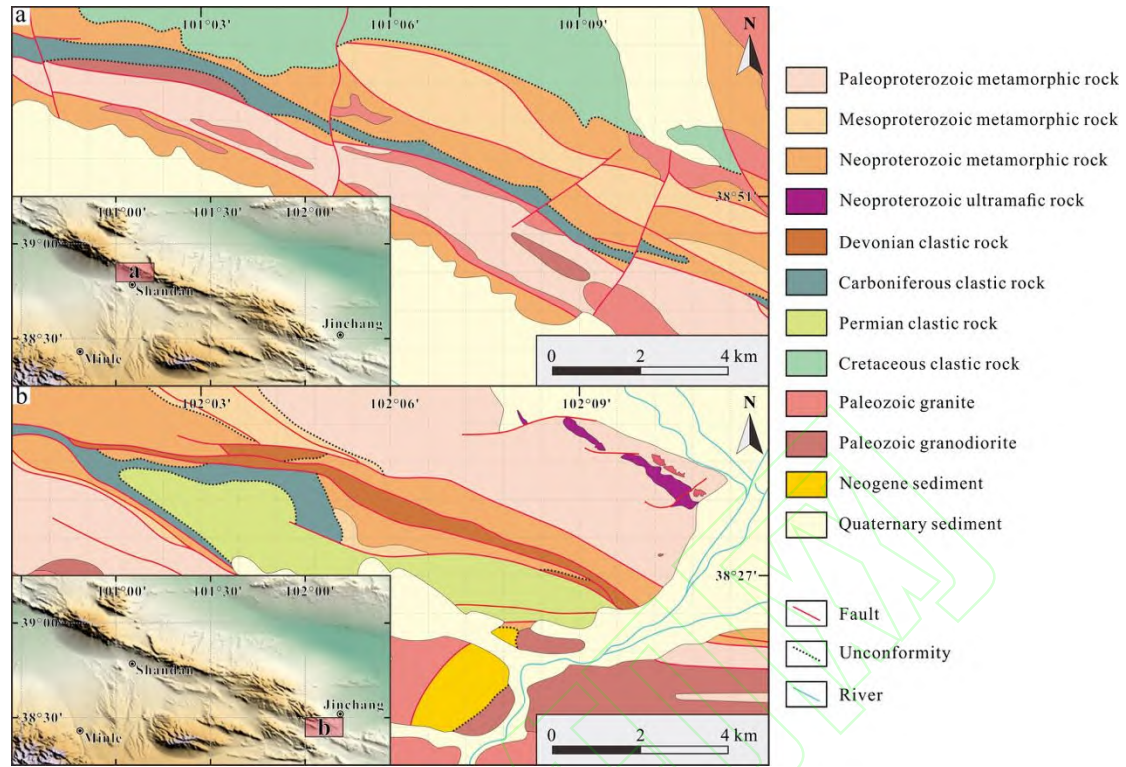


Figure 2. Geological maps of LSSTB showing unconformity between Carboniferous and Mesoproterozoic strata (a) and unconformity between Carboniferous and Meso-/Neoproterozoic strata (b) (after GSBGMR, 1973; GSBGMR, 1995).

1.3 Thermochronological data of the Jinchuan Deposit

Published thermochronological data from the Jinchuan mining area include samples collected from both the mafic-ultramafic intrusions and the surrounding Longshoushan Complex (Table 1). These studies primarily aimed to reconstruct the post-emplacement thermal evolution of the Jinchuan deposit (Fig. 3), as will be discussed in detail in Section 3.1.

Notably, Lei et al. (2023) conducted inverse thermal modeling using AFT data from three granite samples and one gneiss sample in the Longshoushan Complex. Assuming a geothermal gradient of 35 °C/km and applying a Monte Carlo inverse thermal modeling (Ketcham, 2005), they estimated that the region experienced approximately 3.8 km of exhumation since ca. 160 Ma. Based on this, the authors proposed that a significant portion of the Jinchuan ore-bearing intrusion remains buried, given its estimated emplacement depth of 7~10 km (Chen et al., 2009; De Waal et al., 2004; Yang et al., 1998; Yang et al., 1991). However, the Jinchuan ore-bearing intrusion crystallized during

the Neoproterozoic (ca. 830 Ma; Zhang et al., 2010), whereas the inverse thermal modeling history begins from ca. 160 Ma. Comparing exhumation depths across these vastly different time scales may yield misleading conclusions. If we instead consider the modeled near-surface exposure of the intrusion at ca. 15 Ma (Lei et al., 2023), the inferred erosion thickness would be ~1.1 km. This is roughly comparable to the preserved vertical extent of the ore-bearing intrusion in Segment II (~1.2 km, Fig. 1c), suggesting that nearly half of the original volume of the Jinchuan intrusion may have been eroded.

Liu et al. (2023) reconstructed the thermal history of the Jinchuan deposit since ca. 100 Ma using AFT data and Monte Carlo inverse thermal modeling (Ketcham, 2005). However, the sampling locations are closed to the F₈ fault in the mining area, which might lead to an early near-surface exposure timing (ca. 8 Ma) in the inverse thermal modeling result related to the faulting activity. Additionally, Liu et al. (2023) inferred that Segment I experienced ~0.5 km less exhumation thickness than Segment II since ca. 100 Ma, and concluded that Segment I might have greater potential for deep exploration. But it appears inconsistent with field-based geological observations. Segment I has already been exploited as an open-pit mine, and its ore-bearing intrusion shows a significantly reduced residual thickness compared to Segment II (see Fig. 1c). The major orebody in Segment I (No. 24) is also closer to the surface, making it difficult to argue that the preservation of Segment I is superior. Moreover, although AFT and AHe data from Segment I are clearly younger than those from Segment II (Table 1), the number of samples directly collected from the mafic-ultramafic intrusion in Segment I remains limited (Cai et al., 2024; Liu et al., 2023). Therefore, it is currently difficult to quantitatively constrain the thermal and exhumation history of Segment I with high confidence.

To date, the thermal history, the timing of near-surface exposure and exhumation thickness of the Jinchuan ore-bearing intrusion remain poorly constrained in a quantitative sense. This study focuses on Segment II, which provides a more comprehensive thermochronological dataset. By applying inverse thermal modeling, we aim to reconstruct the thermal history and quantitatively constrain the exhumation thickness of the ore-bearing intrusion, thereby offering new insights into the preservation state of the Jinchuan Cu-Ni sulfide deposit.

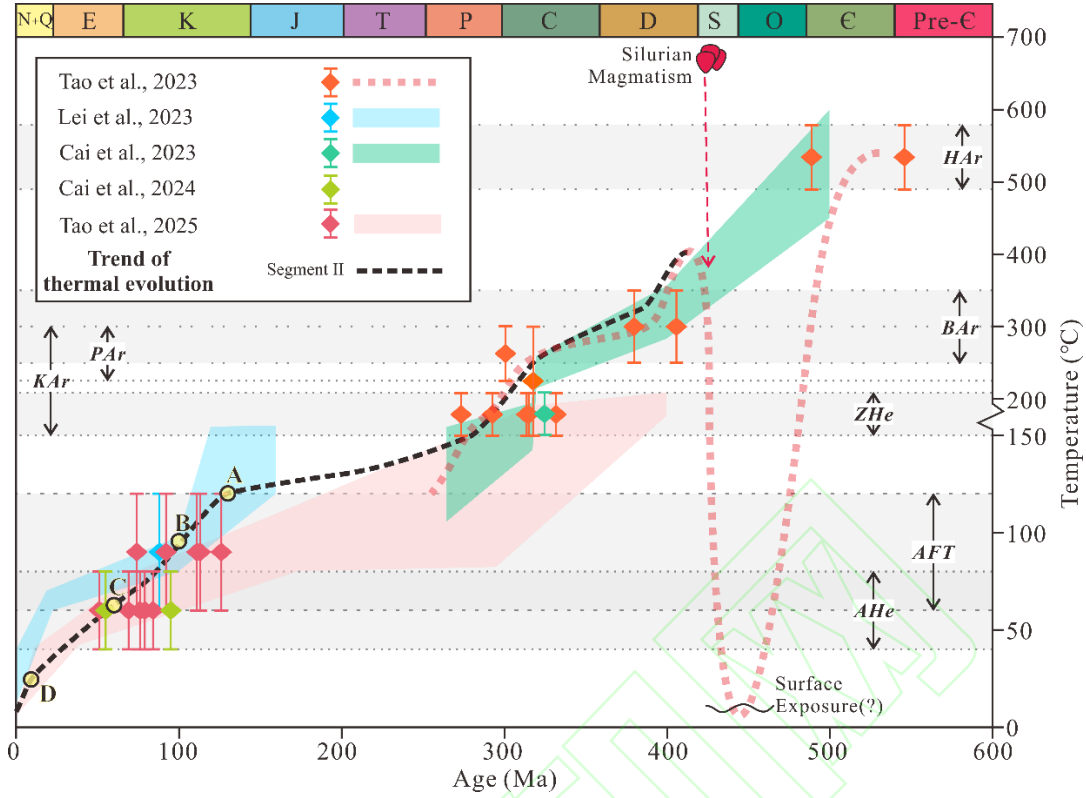


Figure 3. The summary of thermal history of Jinchuan deposit, based on published thermochronological ages and inverse thermal modeling results. HAr: hornblende $^{40}\text{Ar}/^{39}\text{Ar}$ (Harrison, 1982); BAr: biotite $^{40}\text{Ar}/^{39}\text{Ar}$ (Harrison et al., 1985; Scibiorski et al., 2015); PAR: plagioclase $^{40}\text{Ar}/^{39}\text{Ar}$ (Cassata et al., 2009); KAr: K-feldspar $^{40}\text{Ar}/^{39}\text{Ar}$ (Lovera et al., 1989). Translucent yellow dots (A-D) mark key tectonic events discussed in Section 3.1. A: Rapid exhumation of the southern margin of the Alxa Block, possibly related to the Yanshanian movement; B: The Lhasa-Qiangtang collision; C: The initial India-Eurasia collision; D: The last major exhumation event in the Longshoushan, possibly related to activity along the Northern Longshoushan Fault.

2 INVERSE THERMAL MODELING: METHODS AND RESULTS

In this study, 4DTherm v.1.2 software (Fu et al., 2010) is used for inverse thermal modeling of the Jinchuan ore-bearing intrusion. The software employs a two-dimensional explicit finite-difference solution that addresses several processes, including conduction cooling, latent heat of crystallization and fusion, thermal convection within magma bodies, and hydrothermal circulation induced by magma intrusion (Fu et al., 2010; Fu et al., 2004; Leng et al., 2018). The results quantify several parameters related to the dynamic processes of magmatic-hydrothermal cooling, the onset and duration of hydrothermal activity, and the exhumation history of igneous intrusions and associated mineralization (Fu et al., 2010; Fu et al., 2004; McInnes et al., 2005).

In practice, 4DTherm v.1.2 software can simulate the cooling history of an intrusion based on input thermochronological data and calculate the cooling rate at a specific location (Fu et al., 2010). A cooling curve is generated by applying a series of iterative steps (Fu et al., 2010). If the cooling curve passes through all points of thermochronological data, it can be concluded that the curve is correct and successfully defines a plausible cooling history (Fu et al., 2010). The results generated using 4DTherm v.1.2 can be compared and verified with the modeling results of other inverse thermal modeling programs like HeFTy (Ketcham, 2005) to ensure reliability and consistency.

Prior to modeling, several key parameters are defined: (1) **Closure temperature** is assigned based on the thermochronological systems applied, in this study is 180 °C for zircon (U-Th)/He (Guenther et al., 2013; Reiners et al., 2004) and 60 °C for apatite (U-Th)/He (Flowers et al., 2009). Notably, 4DTherm requires a specific closure temperature; (2) **Initial emplacement depth** refers to the vertical distance between the top of the intrusion and the paleo-surface at the time of emplacement; (3) **Geothermal gradient** plays a role in heat transport and exhumation modeling (Fu et al., 2010). However, the long-term geothermal gradient of the Jinchuan deposit remains poorly constrained. Previous studies have typically adopted values between 15~35 °C/km (Lei et al., 2023; Liu et al., 2023; Tao et al., 2023). To evaluate its influence on modeling results, we used multiple geothermal gradients, see 3.2 for details; (4) **Eroded thickness** is defined as the thickness of the intrusion that has been removed due to exhumation, it is generated by 4DTherm v.1.2 once the geometry and initial emplacement depth of the intrusion is set.

The sample 19JC07 is applied in the inverse thermal modeling (Fig. 1b, Tao et al., 2023; Tao et al., 2025). In addition to the thermochronological data and the parameters mentioned above, the geometry of the intrusion should be constrained (Fu et al., 2010). Considering that the original geometry of the Jinchuan ore-bearing intrusion was a nearly horizontal intrusive sheet (Lehmann et al., 2007), the shape of the intrusion is set to be "sheet-like". According to the sampling location, this study constrains the geometry of the intrusion based on the mine section 34 (Duan, 2015) : the width is set to 0.5 km, the residual height is 0.8 km and the dip angle is 65° (Duan, 2015; Li, 2022; Tang et al., 2023). Zircon U-Pb age is from Zhang et al. (2010). The remaining initial parameters are set to default values from 4DTherm v.1.2 (Fu et al., 2010). A complete summary of model inputs and results of inverse thermal modeling are provided in Table 2.

3 DISCUSSIONS

3.1 Thermal History Reconstruction of the Jinchuan Deposit

The thermal evolution of the Jinchuan magmatic system reflects a complex interplay of magmatic activity, sedimentary burial, and tectonic exhumation. The integration of multi-system thermochronological data enables the reconstruction of its cooling history from the late Paleozoic to the Cenozoic.

The earliest recognizable thermal event following emplacement (ca. 830 Ma, Zhang et al., 2010) is a reheating phase that occurred around ca. 400 Ma. Geological mapping shows that the Neoproterozoic strata in the Jinchuan area are directly overlain by Permian units, and that Devonian strata are faulted against Mesoproterozoic units, suggesting that the Jinchuan ore-bearing intrusion had already been brought close to the surface before the Silurian (Tao et al., 2023). This inference is further supported by $^{40}\text{Ar}/^{39}\text{Ar}$ biotite ages from mining segment II (406–380 Ma, corresponding to a closure temperature range of 250–350°C, Harrison et al., 1985; Scibiorski et al., 2015), which suggest reheating to at least 350°C, likely due to Silurian mafic-ultramafic dykes and/or sedimentary burial (Duan, 2015; Lei et al., 2023; Tao et al., 2023). These data constrain the system to a peak temperature of about 400°C around 400 Ma (Tao et al., 2023).

Using the QTQt (Gallagher, 2012), Cai et al. (2023) integrated published $^{40}\text{Ar}/^{39}\text{Ar}$ (Tao et al., 2023) and zircon (U-Th)/He ages (Cai et al., 2023; Tao et al., 2023) from the Jinchuan mining segment II to establish constraints for inverse thermal modeling, identifying two rapid cooling events at ca. 400 Ma and ca. 320 Ma. The latter coincides with the observed zircon (U-Th)/He age distribution, indicating that the system cooled below the ZHe closure temperature by ca. 300 Ma (Cai et al., 2023; Tao et al., 2023). Additional modeling results by Tao et al. (2025) indicate that monotonic cooling continued thereafter.

To explore the long-term cooling trajectory, 4DTherm v.1.2 was used to conduct inverse thermal modeling from both ca. 400 Ma (the reheating event, Duan, 2015; Tao et al., 2023) and ca. 300 Ma (Cai et al., 2023; Tao et al., 2023) starting points. Both scenarios yielded similar average cooling rates of 0.60–0.70°C/Ma. However, according to the time-temperature curve proposed by Tao et al. (2023), the average cooling rates of the Jinchuan deposit since ca. 400 Ma and ca. 300 Ma are 0.75°C/Ma and 0.66°C/Ma, respectively. It seems that inverse thermal modeling using ca. 300 Ma as the starting point is more credible, which constrains the cooling process since when the

Jinchuan ore-bearing intrusion cooled to a temperature below the zircon (U-Th)/He closure temperature range.

The average cooling rate between the time when the Jinchuan deposit passed below the ZHe closure temperature (ca. 300 Ma) and the AFT closure temperature (ca. 130 Ma) is relatively low (Table 3). Two possibilities may explain this pattern: (A) the region underwent tectonic quiescence and peneplanation during the Permian-Triassic; or (B) an unrecognized reheating event occurred between 300 and 130 Ma. The inverse thermal modeling results of Tao et al. (2025) support assumption (A), that is, the Longshoushan thrust belt, including the Jinchuan deposit, experienced a stage of tectonic quiescence and peneplanation during the Permian-Triassic, which is consistent with the shallowly exhumed basement units across the region (Tao et al., 2025). For assumption (B), there is no significant magmatism in where the Jinchuan deposit is located during ca. 300 Ma to ca. 130 Ma (Gao, 2009). And given that Jurassic and Cretaceous strata are not presented in the Jinchuan mining area (GSBGMR, 1995), it is unlikely that a significant reheating event caused by sedimentary burial.

The rapid cooling of the Jinchuan deposit around 130 Ma coincides with the regional uplift and exhumation of the southern margin of the Alxa Block, likely related to the late phase of the Yanshanian movement (GSBGMR, 1995). The absence of Late Cretaceous strata in the Jinchuan area and within the Alax Block further supports significant denudation and exhumation during this period (Zhao, Y. F. et al., 2023). The cooling rate of Jinchuan deposit suddenly increases as early as ca. 130 Ma (Fig. 3), indicating that Jinchuan ore-bearing intrusion was likely exhumed to the near-surface and began to be eroded. If the assumption is true, the inverse thermal modeling results suggesting that the near-surface exposure of Jinchuan ore-bearing intrusion is no earlier than ca. 130 Ma is believable.

Published AFT ages (126~73.6 Ma) and apatite (U-Th)/He ages (94.8~50.5 Ma) show overlap between 95 and 74 Ma, suggesting the rapid cooling of the Jinchuan deposit through the 60~80°C range (Flowers et al., 2009; Tagami and O'Sullivan, 2005) during the Late Cretaceous. Notably, this rapid cooling interval (ca. 95~74 Ma) lies between two major cooling events inferred by Cai et al. (2024) from inverse thermal modeling (Monte Carlo method, Gallagher, 2012) of AHe data: ca. 115~95 Ma and ca. 65~55 Ma. The former has been attributed to the Lhasa-Qiangtang collision, while the latter is interpreted as a response to the initial India-Asia collision (Cai et al., 2024).

Therefore, the transition of the Jinchuan deposit from the AFT to the AHe closure temperature range may have been predominantly driven by these two major tectonic events, which are both related to the closure of the Paleo-Tethys Ocean (Lei et al., 2023). Liu et al. (2023) suggest that the Jinchuan deposit was slowly exhumed during ca. 100 Ma to ca. 53 Ma based on the inverse thermal modeling results of AFT data (Monte Carlo method). However, based on the distribution of apatite fission-track and apatite (U-Th)/He data, it is possible that the current exposed surface of the Jinchuan ore-bearing intrusion, after leaving the closure temperature range of AFT at ca. 74 Ma at the earliest, was mainly influenced by the initial India-Eurasia collision (Cai et al., 2024) until leaving the closure temperature range of apatite (U-Th)/He at ca. 50 Ma.

The final rapid cooling phase likely began no later than the early Pliocene. Lei et al. (2023) date this event to ca. 15 Ma, while Liu et al. (2023) argue for ca. 8 Ma, which coincides with the latest timing when the accelerated crustal shortening extended into the northeastern Tibetan Plateau during the Miocene (23~8 Ma, Cai et al., 2024). Tao et al. (2025), using apatite fission-track and (U-Th)/He analysis, coupled with 2D thermal-kinematic modeling, suggest the last major exhumation occurred between 10 and 5 Ma, related to activity along the Northern Longshoushan Fault (N-LSSF), which is not associated with the initial exhumation of the ore-bearing intrusion.

3.2 Near-Surface Exposure of Jinchuan Deposit

When the geothermal gradient is set to 36°C/km, the initial emplacement depth is set to 3200 m, and the eroded thickness is constrained to ~1860 m, the obtained modeled age of the sample is the closest to the actual age (ZHe: 281 Ma; AHe: 79.1 Ma), and the time of near-surface exposure of Jinchuan ore-bearing intrusion is constrained to be ca. 106 Ma (Fig. 4). If "the difference between the modeled age and the actual age is less than 1 Ma for apatite (U-Th)/He, and less than 2 Ma for zircon (U-Th)/He" is taken as the criterion for the credibility of the inverse thermal modeling results, then the inverse thermal modeling results reflect that the Jinchuan ore-bearing intrusion was exposed to the near-surface during ca. 87 to ca. 121 Ma (Table 2). During this period, the samples were cooled to a range of the overlap of the closure temperature ranges of AFT and apatite (U-Th)/He.

For successful inverse thermal modeling results from the 4DTherm v.1.2, with a constant preset geothermal gradient, the larger the initial emplacement depth of the intrusion, the smaller the eroded thickness and the later the near-surface exposure of the intrusion. While with a constant preset initial

emplacement depth of the intrusion, the higher the geothermal gradient, the smaller the eroded thickness and the later the near-surface exposure of the intrusion (Fu et al., 2010). For example, the inverse thermal modeling result No.18 in Table 2 also meets the criterion, and the timing of near-surface exposure is modeled to be ca. 9 Ma, which seems to support that the Jinchuan ore-bearing intrusion was exposed to the near-surface in the Miocene (Lei et al., 2023). However, the present geothermal gradient in the Jinchuan mining area is $\sim 20^{\circ}\text{C}/\text{km}$ (from measured data at the mining area), the eroded thickness of the Jinchuan ore-bearing intrusion can be calculated to be ~ 900 m, which is not conform to the modeled one. Therefore, this result is not highly credible. Considering the results of inverse thermal modeling and the thermal history reconstruction, the Jinchuan ore-bearing intrusion was more likely exposed to the near-surface at ca. 90 Ma, at the timing when the Jinchuan ore-bearing intrusion cooled to the bottom of the closure temperature of AHe (Cai et al., 2024; Tao et al., 2025). Once the thermal history has been sufficiently and accurately characterized, the exhumation can be calculated based on Fig 8a in Tao et al. (2025) using the approach described in Yuan et al. (2011) and Yuan (2016).

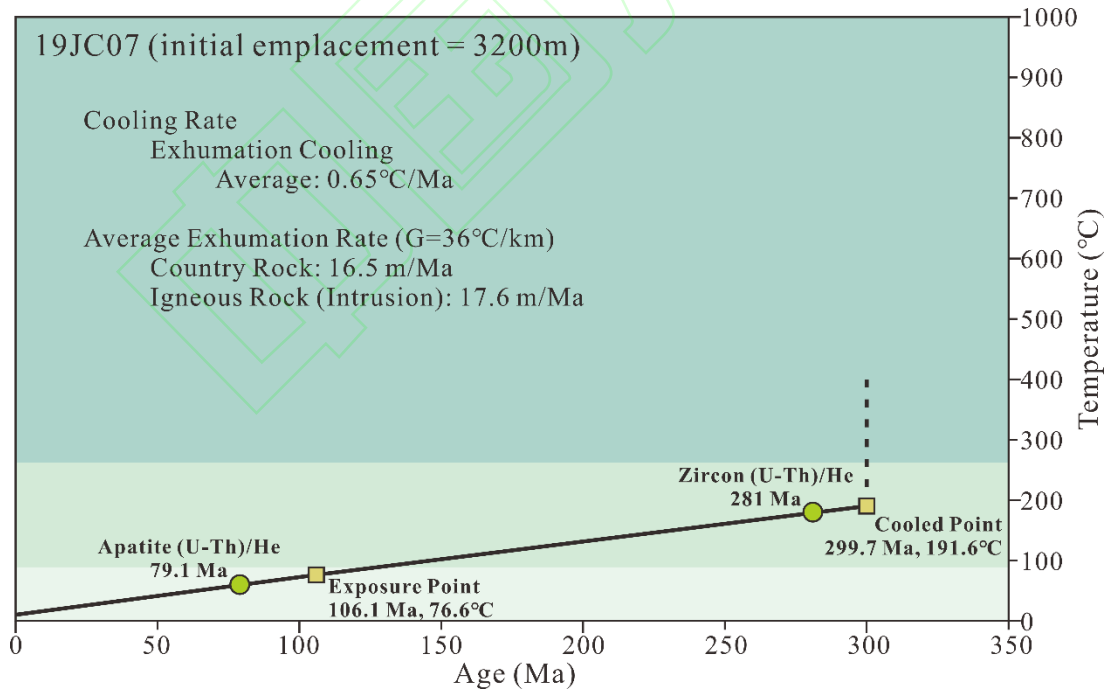


Figure 4. 2D explicit finite inverse thermal modeling results of the exhumation path and surface exposure timing of Jinchuan deposit.

3.3 Exhumation History and Preservation Status

Based on the result of quantitative calculation, this study summarizes the exhumation history of the Jinchuan deposit since ca. 300 Ma (Fig. 5). Geothermal gradients used to calculate the exhumation are based on Ren (1998) and Ren and Zhao (2001), the present surface temperature is set to 12 °C. (1) At ca. 300 Ma, the current samples of the Jinchuan mining area are located 2.7~6.2 km below the surface. During ca. 300 Ma to ca. 200 Ma, the Jinchuan ore-bearing intrusion was exhumed to the surface for 0.2~2.5 km. Assuming the elevation of the Jinchuan mining area is about -0.2 km at ca. 300 Ma (shallow marine environment, Gao, 2009) and ~0.5 km at ca. 200 Ma (mountainous terrain environment, GSBGMR, 1995), then the uplift during this period is about 0.9~3.2 km (Fig. 5a), which is likely be a response to subduction and closure of both the Paleo-Asian and Paleo-Tethys Oceans during the Permian to Triassic (Tao et al., 2025); (2) from ca. 200 Ma to ca. 130 Ma, the cumulative exhumation of the Jinchuan deposit is 0.3~1.4 km. Assuming that the Jinchuan mining area is at an elevation of ~0.5 km at ca. 130 Ma, the uplift during this period is 0.3~1.4 km; (3) from ca. 130 Ma to ca. 90 Ma, the cumulative exhumation of the Jinchuan deposit is 0.4~1.5 km. Assuming the elevation of Jinchuan mining area is ~0.5 km at ca. 90 Ma, the uplift during this period is 0.4~1.5 km (Fig. 5b); (4) since the near-surface exposure of the Jinchuan ore-bearing intrusion in the ca. 90 Ma to ca. 5 Ma, which is the latest time of the N-LSSF movement (Tao et al., 2025), the cumulative exhumation of the Jinchuan ore-bearing intrusion is 1.8~3.7 km, and the uplift during this period is 2.9~4.8 km as the current average elevation of the Jinchuan mining area is 1.6 km (Fig. 5c, d, GSBGMR, 1995).

The satisfactory outcome of the inverse thermal modeling using 4DTherm v.1.2 indicates that the predefined emplacement depth and erosion thickness of the intrusion are geologically plausible (Fu et al., 2010). Although other combinations of emplacement depth and eroded thickness could also yield acceptable fits within the model constraints, the parameter space for the Jinchuan intrusion appears to be relatively limited (Fu et al., 2010). In the best-fitting model, the adopted geothermal gradient (36 °C/km) is notably higher than the present-day gradient in the Jinchuan mining area (20 °C/km), implying that the modeled erosion thickness may be underestimated. Nevertheless, the modeled eroded thickness exceeds the current residual height of the Jinchuan ore-bearing intrusion (600-1200 m, Duan, 2015; Gao et al., 2009; Tang et al., 2009), suggesting that more than half of the original intrusion volume has been removed. If the exhumation thickness since ca. 90 Ma is taken

to represent the total erosion affecting the intrusion (Table 2), then at least 67% of its volume has been eroded (Duan, 2015; Gao et al., 2009; Tang et al., 2009). Given that some orebodies are currently exposed at the surface, this post-exposure exhumation likely had a significant influence on the preservation state of the Jinchuan deposit. However, in the case of Segment II of the Jinchuan mining area, the removal of the upper to middle portions of the intrusion may facilitate deeper exploration by reducing the amount of overburden.

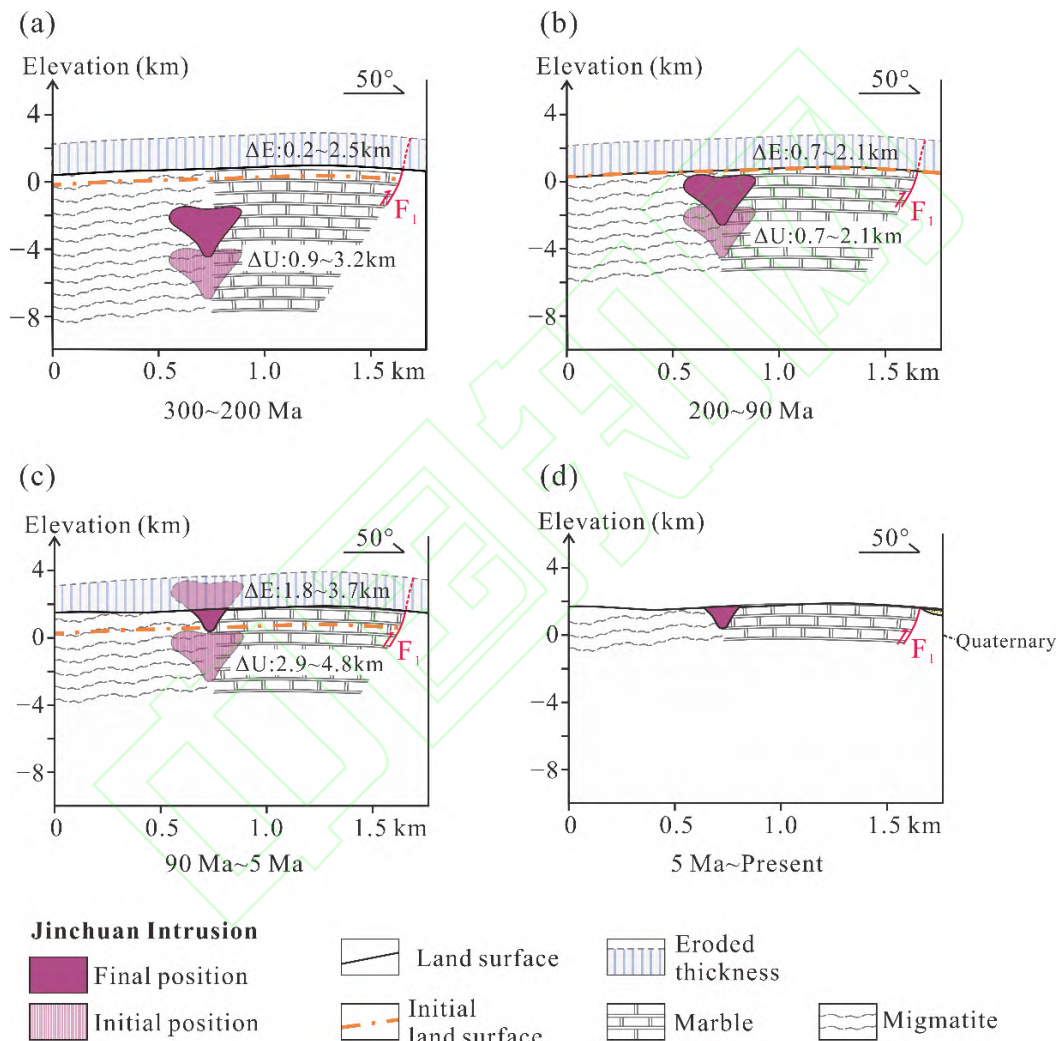


Figure 5. Uplift and exhumation history of the Jinchuan mining area since ca. 300 Ma. (a) 300~200 Ma; (b) 200~90 Ma; (c) 90~5 Ma; (d) 5 Ma~ Present. The size, pattern and symbol of the rocks are for illustration purposes only. F₁: N-LSSF.

4 CONCLUSIONS

This study compiles and reconstructs the thermal history of the Jinchuan Cu-Ni sulfide deposit

with low-temperature thermochronological ages (zircon (U-Th)/He, apatite (U-Th)/He, apatite fission track), and Monte Carlo inverse thermal modeling results. Combined with 2D explicit finite difference inverse thermal modeling, we decode the timing of near-surface exposure for the Jinchuan ore-bearing intrusion and quantitatively constrain its post-exposure exhumation thickness. The above sheds light on, the preservation status of the giant Jinchuan deposit. Our main conclusions are as below.

(1) The Jinchuan deposit was cooled to a temperature below the zircon (U-Th)/He closure temperature range at ca. 300 Ma; the region geology and absence of magmatic record may rule out the possibility of nonmonotonic cooling process in following periods;

(2) The Jinchuan ore-bearing intrusion is likely to have been exposed to the near-surface in the Late Cretaceous (ca. 90 Ma), corresponding to when the samples are cooled to a temperature in the intersection of closure temperature ranges of apatite fission-track and apatite (U-Th)/He;

(3) The exhumation since near-surface exposure is approximately twice the current maximum thickness of the residual Jinchuan ore-bearing intrusion (1.2 km), indicating that prolonged post-mineralization exhumation facilitates shallow exhumation of this ancient pre-Cambrian deposit and ultimately leads to exposure and partial erosion of the Jinchuan deposit.

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Table 1. Published thermochronological ages in Jinchuan mining area

Sample	Lithology	Latitude (N)	Longitude (E)	Elevation (m)	Analytical technique	Age (Ma, $\pm 1\sigma$)	Reference	Corresponding area
19JC02	amphibolite	38°28'45.7"	102°10'17.2"	1620	Hbl Ar	555.6 $\pm$ 4.9	Tao et al. (2023)	Segment II
19JC11-1	migmatitic granite	38°28'34.3"	102°10'7.3"	1650	Hbl Ar	488.9 $\pm$ 3.9	Tao et al. (2023)	
19JC02	amphibolite	38°28'45.7"	102°10'17.2"	1620	Bi Ar	379.6 $\pm$ 2.3	Tao et al. (2023)	
19JC11-1	migmatitic granite	38°28'34.3"	102°10'7.3"	1650	Bi Ar	406.3 $\pm$ 2.3	Tao et al. (2023)	
19JC02	amphibolite	38°28'45.7"	102°10'17.2"	1620	Kfs Ar	318.2 $\pm$ 2.1	Tao et al. (2023)	
19JC10	migmatitic granite	38°28'34.3"	102°10'7.3"	1650	Pl Ar	301.4 $\pm$ 2.1	Tao et al. (2023)	
19JC01	migmatitic granite	38°28'46.4"	102°10'16.9"	1620	ZHe	313 $\pm$ 15	Tao et al. (2023)	
19JC02	amphibolite	38°28'45.7"	102°10'17.2"	1620	ZHe	284 $\pm$ 11	Tao et al. (2023)	
19JC03	granite	38°28'44.0"	102°10'15.8"	1620	ZHe	272 $\pm$ 15	Tao et al. (2023)	
19JC07	pyroxenite	38°28'25.9"	102°10'37.8"	1630	ZHe	281 $\pm$ 16	Tao et al. (2023)	
19JC10	migmatitic granite	38°28'34.3"	102°10'7.3"	1650	ZHe	313 $\pm$ 14	Tao et al. (2023)	
19JC11-1	migmatitic granite	38°28'34.3"	102°10'7.3"	1650	ZHe	333.8 $\pm$ 7.4	Tao et al. (2023)	
JC-20-T	gabbro	38°28'36.5"	102°10'29.4"	1644	ZHe	324.66 $\pm$ 9.36	Cai et al. (2023)	
JC-32	granite	38°28'48.0"	102°10'16.0"	1597	AFT	88.4 $\pm$ 4.4	Lei et al. (2023)	
19JC01	migmatitic granite	38°28'46.4"	102°10'16.9"	1620	AFT	126 $\pm$ 7.2	Tao et al. (2025)	
19JC02	amphibolite	38°28'45.7"	102°10'17.2"	1620	AFT	110.9 $\pm$ 4.4	Tao et al. (2025)	
19JC03	granite	38°28'44.0"	102°10'15.8"	1620	AFT	73.6 $\pm$ 4.7	Tao et al. (2025)	
19JC07	pyroxenite	38°28'25.9"	102°10'37.8"	1630	AFT	91.8 $\pm$ 5.1	Tao et al. (2025)	
19JC11-1	migmatitic granite	38°28'34.3"	102°10'7.3"	1650	AFT	112.7 $\pm$ 6.1	Tao et al. (2025)	
19JC01	migmatitic granite	38°28'46.4"	102°10'16.9"	1620	AHe	84.2 $\pm$ 3.4	Tao et al. (2025)	
19JC02	amphibolite	38°28'45.7"	102°10'17.2"	1620	AHe	76 $\pm$ 4.4	Tao et al. (2025)	
19JC03	granite	38°28'44.0"	102°10'15.8"	1620	AHe	68.9 $\pm$ 3.3	Tao et al. (2025)	

19JC07	pyroxenite	38°28'25.9"	102°10'37.8"	1630	AHe	79.1±4	Tao et al. (2025)	Segment II
19JC11-1	migmatitic granite	38°28'34.3"	102°10'7.3"	1650	AHe	50.5±5.2	Tao et al. (2025)	
JC-3-Gb	gabbro	38°28'34.4"	102°10'26.0"	1582	AHe	54.85±0.28	Cai et al. (2024)	
JC-20-T	gabbro	38°28'36.5"	102°10'29.4"	1644	AHe	94.75±6.12	Cai et al. (2024)	
JC-37	granite	38°28'46.0"	102°10'59.0"	1589	AFT	81.4±5.5	Lei et al. (2023)	Segment II (offset from the ore-bearing intrusion)
JC-36	granite	38°28'42.0"	102°09'25.0"	1659	AFT	82.4±6.4	Lei et al. (2023)	
JC20-3	migmatite	38°28'59.0"	102°09'30.0"	1627	AFT	40.8±5.4	Liu et al. (2023)	
JC20-4	migmatite	38°28'57.0"	102°09'28.0"	1630	AFT	30.8±10.4	Liu et al. (2023)	
LSLKB-1453	granite	38°29'19.8"	102°08'59.5"	1453	ZHe	310.55±8.18	Cai et al. (2023)	Segment I
LSLKB-1453	granite	38°29'19.8"	102°08'59.5"	1453	AHe	42.01±7.69	Cai et al. (2024)	
LS-3	pyroxene peridotite	38°29'23.2"	102°08'49.8"	1482	AHe	22.88±1.65	Cai et al. (2024)	
LS-4	pyroxene peridotite	38°29'23.6"	102°08'48.0"	1493	AHe	29.64±0.64	Cai et al. (2024)	
LSLK-1560	granite	38°29'28.8"	102°08'40.7"	1560	ZHe	280.6±1.11	Cai et al. (2023)	Segment I (offset from the ore-bearing intrusion)
LSLK-1560	granite	38°29'28.8"	102°08'40.7"	1560	AHe	31.05±14.17	Cai et al. (2024)	
JC-29	granite	38°29'31.0"	102°08'33.0"	1620	AFT	71.1±4.8	Lei et al. (2023)	
JC20-15	migmatite	38°29'4.0"	102°08'41.0"	1585	AFT	50.9±4.6	Liu et al. (2023)	

Hbl Ar: hornblende $^{40}\text{Ar}/^{39}\text{Ar}$ (Harrison, 1982); Bi Ar: biotite $^{40}\text{Ar}/^{39}\text{Ar}$ (Harrison et al., 1985; Scibiorski et al., 2015); Pl Ar: plagioclase $^{40}\text{Ar}/^{39}\text{Ar}$ (Cassata et al., 2009); Kfs Ar: K-feldspar $^{40}\text{Ar}/^{39}\text{Ar}$ (Lovera et al., 1989). ZHe: zircon (U-Th)/He (Guenther et al., 2013; Reiners et al., 2004); AFT: apatite fission-track (Tagami and O'Sullivan, 2005); AHe: apatite (U-Th)/He (Flowers et al., 2009).

Table 2 Input data and output results of inverse thermal modelling by 4DTherm v.1.2

No.	Geothermal gradient (°C/km)	Initial emplacement depth (m)	Eroded thickness (m)	Modeled average cooling rate (°C/Ma)	Modeled exposure point		Modeled age (Ma)	
					Time (Ma)	Temperature (°C)	AHe	ZHe
1	70	2500	181.90	0.64	16.39	23.4	77.91	279.81
2	60	2900	190.75	0.63	14.22	20.1	81.28	283.18
3	50	3000	703.15	0.64	54.12	46.4	77.07	278.97
4	44	3000	1120.29	0.63	77.89	58.4	80.60	282.50
5	42	3000	1361.08	0.64	90.42	68.1	77.55	278.63
6	42	2900	1457.23	0.64	96.81	71.9	77.93	278.63
7	42	2800	1553.37	0.64	103.19	75.7	78.31	278.63
8	40	4000	604.52	0.64	35.65	34.9	77.91	279.81
9	40	3500	1097.82	0.64	68.84	54.6	77.91	279.81
10	40	3200	1370.73	0.64	86.73	65.2	78.44	279.81
11	40	3100	1443.90	0.64	91.35	68.0	78.71	279.81
12	40	3000	1592.69	0.64	100.77	73.6	79.28	279.81
13	40	2900	1682.98	0.64	106.49	77.0	79.62	279.81
14	38	3100	1720.96	0.65	103.44	75.6	78.80	279.14
15	38	3000	1830.19	0.65	110.11	79.5	79.20	279.14
16	38	2900	1926.65	0.65	115.81	82.9	79.54	279.14
17	38	2800	2436.45	0.65	120.83	85.9	79.84	279.14
18	36	4800	228.61	0.65	8.80	18.7	78.25	280.15
19	36	3300	1774.55	0.65	101.05	73.6	79.61	280.15
20	36	3200	1863.56	0.65	106.12	76.6	79.12	280.15
21	36	3100	1978.96	0.65	112.69	80.5	80.30	280.15
22	35	3200	2016.40	0.66	111.64	80.0	79.93	279.81

23	35	3000	2227.51	0.65	123.32	88.7	77.85	276.87
24	35	2800	2436.45	0.66	133.97	95.0	78.48	276.87
25	35	2700	2535.46	0.66	140.37	98.9	78.87	276.87
26	32	3000	2735.46	0.67	138.46	98.0	78.36	276.45
27	29	3000	3297.88	0.68	151.29	105.8	78.89	276.19
28	26	3000	4011.87	0.70	165.00	114.4	79.00	275.44
29	25	3000	4281.38	0.87	169.32	117.3	78.62	274.76
30	24	3000	4577.97	0.90	173.81	119.8	79.21	275.10
31	20	3100	6091.36	1.05	192.71	132.2	78.44	273.08

The input data includes “geothermal gradient” and “initial emplacement depth”, while the rest are output results. *No.* means the serial number of inverse thermal modeling records.

Table 3 Average cooling rates according to different thermochronological systems

Sample ID	Average Cooling Rate (°C/Ma)		
	ZHe-AFT	AFT-AHe	AHe-Exposed
19JC01	0.48	0.72	0.57
19JC02	0.52	0.86	0.63
19JC03	0.45	6.38	0.70
19JC07	0.48	3.94	0.61
19JC11-1	0.41	0.48	0.95

ZHe: zircon (U-Th)/He, with the closure temperature taken as 180°C (Guenther et al., 2013; Reiners et al., 2004);

AFT: apatite fission-track, with the closure temperature taken as 90°C (Tagami and O'Sullivan, 2005); AHe: apatite (U-Th)/He, with the closure temperature taken as 60°C (Flowers et al., 2009).